Analytical Chemistry Cumulative Examination
February 27, 2004

1. A peak in the UV-VIS part of the spectrum has a maximum at 582 nm and a full width half maximum (FWHH) of 100 nm. (a) What is the energy of the transition based on its λ_{max}? (b) To what position in wavenumbers do 532 and 632 nm (the FWHH boundaries) correspond? (c) What is the bandwidth of the peak in wavenumbers? (d) What process is most likely to give rise to this transition? (e) What is the frequency of light that corresponds to the maximum absorbance? (30 points)

2. A spectrometric assay is designed to work with final concentrations of 0.005 to 0.020 mg/L of nicotine. If a 500 ml sample of urine is used and all of the nicotine present is concentrated into a final solution volume of 10.00 mL, what is the minimum concentration of nicotine that can by detected in urine by the assay? (10 points)

3. Consider that a single sample was analyzed using two different spectrometric instruments. One has 1.0% stray light at the assay wavelength, the other has effectively no stray light. What relative error in concentration would the 1.0% stray radiation cause if, on the better instrument, various samples had T=0.40, T=0.10, and T=0.05? (30 points)

4. A spectrophotometric assay was to be carried out on a compound with ε=5000. The stock solution of standard had been made with a concentration of 0.0203 M. Given a pathlength of 1.000 cm and the fact that the final absorbance should be around 0.4 to 0.5, by what factor should the stock solution be diluted? (10 points)

5. Explain, with the help of a suitable clearly-labeled diagram, the difference between molecular fluorescence and phosphorescence. Your explanation should include discussion of the time frames involved. (20 points)

Possibly useful constants:

$h=6.62 \times 10^{-34}$ joule sec $c=3.00 \times 10^{8}$ m/sec Avogadro number 6.02×10^{23}